

Course code: CE 241

Course Name: Environmental Engineering I

Pre-requisite: None

Unit 1: Ecology, Environment and Human Health - their linkages; **(4 Lectures)**

Unit 2: Environmental pollution : its origin. Sources of water : surface, ground and others. **(4 Lectures)**

Unit 3: Ground water engineering : Basic aquifer parameters and hydraulics, well system, ground water pollution and control. **(4 Lectures)**

Unit 4: Water quality parameters : Their role and standards. Water demand for various purposes; Estimation of population growth. **(4 Lectures)**

Unit 5: Basic hydraulics (including pumping) related to water supply & wastewater collection, pipe network design. **(4 Lectures)**

Unit 6: Sewerage system : Types of collection & disposal system, design approach to sanitary & storm sewers, sewer appurtenances, street inlets, inverted siphon. **(3 Lectures)**

Unit 7: House plumbing for water supply & wastewater drainage. **(3 Lectures)**

Suggested Books:

1. Peavy and Rowe, 'Environmental Engg.'
2. B.C. Punmia, 'Water Supply & Wastewater Engg.'
3. G.S. Birdie, 'Water Supply & Sanitary Engg.'
4. P.D. Sharma, 'Ecology and Environment'